Comparison of Circuit Switching and Packet Switching

Comparison of Circuit Switching and Packet Switching

S.No	Parameter	Circuit switching Network	Packet switching Network
1	Path	In circuit switched network a dedicated path is created between two points by setting the switches.	In packet switched network no dedicated path is created between two points. Only the virtual circuit exists.
2	Store and forward transmission	In circuit switching there is no concept of store and forward transmission.	In virtual packet switched network, each node may store incoming packets and forward them after use.
3	Dedicated	The links that make a path in circuit switched network are dedicated and cannot be used for other connections.	In the virtual circuit network, links that make a route can be dedicated with other connections.
4	Availability of Bandwidth	In circuit switching, bandwidth is fixed because it is reserved in advance.	In the virtual circuit network, require bandwidth is dynamic because it can be released as it is needed.
5	The route followed by packets	The route followed by packets is always the same.	The route followed by packets is may or may not be different.
6	Call setup	An in-circuit switching call setup is required.	In packet switching, call setup is not required.
7	Congestion	In circuit switching, congestion can occur at set up time.	In packet switching, congestion can occur on every packet.
8	Wastage of Bandwidth	In circuit switching, bandwidth is fixed, unused bandwidth on an allocated circuit is wasted.	Other packets from an unrelated source may utilize unused bandwidth.

Comparison of Circuit Switching and Packet Switching

S.No	Parameter	Circuit switching Network	Packet switching Network
9	Charging	In circuit switching, users are charged based on time and the basis of distance.	In packet switching, users are charged based on time and number of bytes carried & not based on distance.
10	Application	Telephone network for bidirectional, real time transfer of voice signal.	Internet for datagram and reliable stream service between computers.
11	Layers	Circuit-switched network is implemented at the physical layer.	A virtual circuit network is implemented at the data link and a network layer.
12	Reliability	Circuit-switched is highly reliable.	In packet switching, low reliability, subject to congestion.
13	Overhead bits	In Circuit-switched network, no overhead bits after call setup.	In packet switching, Overhead bits in each packet.
14	Technologies or types	Circuit switching using two technologies: Time Division Switching & Space Division Switching	Packet Switching using two technologies Datagram circuit approach & Virtual circuit Approach
15	Installation Cost	Circuit switching's initial cost is low.	Packet switching networks have high installation costs.
16	Protocols	Circuit switching requires simple protocols for delivery.	Packet switching requires complex protocols for delivery.

Comparison of Circuit Switching and Packet Switching

S.No	Parameter	Circuit switching Network	Packet switching Network
17	Addressing scheme	In Circuit switching, Hierarchical numbering plan scheme is used.	In Packet switching, Hierarchical address space is used.
18	End Terminal	In this telephone and modem is used as end terminal.	In this computer is used as end terminal.
19	Information type	In this information type is Analog voice or PCM digital voices.	In this information type is binary information.
20	Multiplexing scheme	In circuit switching, circuit multiplexing is used.	In packet switching, packet multiplexing shared media access network in used.
21	Routing Scheme	In circuit switching, route selecting during set up.	In packet switching, each packet is routed independently.
17	Addressing scheme	In Circuit switching, Hierarchical numbering plan scheme is used.	In Packet switching, Hierarchical address space is used.
18	End Terminal	In this telephone and modem is used as end terminal.	In this computer is used as end terminal.
19	Information type	In this information type is Analog voice or PCM digital voices.	In this information type is binary information.

Difference between Virtual Circuits and Datagram Networks

Key	Virtual Circuits	Datagram Networks
Definition	Virtual Circuit is a connection-oriented service in which there is an implementation of resources like buffers, CPU, bandwidth, etc., used by virtual circuit for a data transfer session.	Datagram networks are a type of connectionless service where no such resources are required for data transmission.
Path	In Virtual circuits, as all the resources and bandwidth get reserved before the transmission, the path which is utilized or followed by first data packet would get fixed and all other data packets will use the same path and consume same resources.	In a Datagram network, the path is not fixed as data packets are free to decide the path on any intermediate router on the go by dynamically changing routing tables on routers.
Header	As there is same path followed by all the data packets, a common and same header is being used by all the packets.	Different headers with information of other data packet is being used in Datagram network.
Complexity	Virtual Circuit is less complex as compared to that of Datagram network.	Datagram network are more complex as compared to Virtual circuit.
Reliability	Due to fixed path and assurance of fixed resources, Virtual Circuits are more reliable for data transmission as compared to Datagram network.	Datagram networks, due to their dynamic resource allocation and dynamic path, are more errorprone and less reliable than Virtual circuits.
Example and Cost	Virtual circuits are costlier in installation and maintenance. They are widely used by ATM (Asynchronous Transfer Mode) Network, which is used for the Telephone calls.	Datagram networks are cheaper as compared to the Virtual Circuits. They are mainly used by IP network, which is used for Data services like Internet.